

Overview

SU-BBC is a cave survey data reduction program for the BBC micro. Survey data are kept on one or more disc (or network) files, and the program produces a list of station coordinates and a number of files which may be used to generate graphical displays.

Features:

Alphanumeric survey station names of up to six characters.

Up to 1500 survey legs in any one run.

Input data may be split over several files for ease of editing.

Compass and clino calibration may be changed as often as required.

Handles both conventional tape/compass/clino surveys, and underwater tape/compass/depth-gauge surveys for cave divers.

Automatic conversion for old survey data in feet.

Survey coordinate range up to plus or minus 20 Kilometres.

Multiple origin definitions for plotting disconnected surveys.

Configuration

The program requires a BBC micro or Master microcomputer with a random access filing system (disc, advanced disc, network etc.). The current version will not work if more than one such filing system is installed, nor on a system with a second processor switched on, though it is intended to fix both of these failings in an early new release.

A text editor is needed to generate the input data for the program, and any editor or wordprocessor which can separate lines by carriage return and does not insert weird characters outside user control will do the job: *Edit, View or Wordwise are all suitable and probably other products too. Again, new versions of the program later this year will support text prepared on editors which use linefeed rather than carriage return, such as Twin.

Entering survey data

In general terms, a cave survey comprises a string of sets of three numbers - a distance, a compass bearing and a clinometer reading. The basic function of a data reduction program is to convert these numbers to (x,y,z) vectors and then add them up to form cartesian coordinates for each survey station. The complexities arise from the need to calibrate instruments, which may vary from one part of a survey to another, perhaps done at a later date, from the fact that caves are seldom single passages, but possess junctions, side passages, and loops, which require a survey to have a method for referring back to previously surveyed points, and from the unhelpful habits of cavers taking backsights, or starting from the end of a new passage and working towards the known survey. Problems can also arise from using existing survey data which may be in different units, and be referred to different origins or coordinate systems. Hence, in addition to the three basic numbers referred to above, we must also input a unique name or number for the survey station at each end of the survey leg, and a certain amount of supplementary

information, such as compass and clino corrections, coordinates of the origin(s) of the survey, and changes of measurement unit. It is also possible to enter data in a format suitable for cave diving surveys, where depth gauge readings are used instead of clinometer readings. A survey may cover more than one cave system, where the starting points are known from high-grade mapping, so a facility is provided to define additional origins. Input data files may contain useful commentary, which is currently reflected onto the screen during the first stage of processing, but not carried through to the final coordinate printout. User feedback on the desirability of this would be appreciated.

To cope with these diverse needs, SU is driven by a series of keywords in the input file. For fairly simple surveys, however, the user need not be concerned with the majority of these keywords, except to ensure that she does not use the keywords as names for her survey stations.

The keywords in use in the current version are as follows:

ORIGIN, CHANGE, DIVING, NORMAL, INFEET, METRIC, FILE, END

A number of keywords are reserved for future expansion, and the user should avoid naming his stations with anything that might be a keyword in the future (sorry about this rather open-ended restriction, but it is unlikely that the surveyor would generate clashes, so don't worry about it). In particular, the following are definitely potential keywords for future versions:

SYSTEM, FIXED, FREE, REMARK, CLOSE, FORMAT, OSCILL, ... NOTES

Survey data is currently input in a fixed format, mainly to ensure data compatibility with the mainframe versions of SU. This format of input will always be supported, but a rather more 'relaxed' free-format input will be supported in future versions of the program. Some fields are also used for the input of control information - ie. keywords and associated data. To ensure that the program never reads a line in an incorrect format, any data which will not fit into the standard layout is placed on a line following the appropriate keyword, which does fit in. The only exceptions to this are the first line of input which contains a title for the survey in the first 48 columns, and no other information, and the second line which contains the coordinates of the initial origin.

The following description of the data input format is inevitably a little involved, and is best read in conjunction with an example of some real input data.

The standard format for data lines comprises two six-character alphanumeric fields, followed by four six-digit real-numeric fields, and finally, a variable length alphanumeric field for commentary. The input formats of the first two lines, and the interpretations of the standard input format for each keyword are as follows. In all cases lower case x indicates that a field is unused. Although a decimal point is indicated in the numeric fields, this may be in any position in the field, or may be omitted altogether. If a numeric field is left blank, it will take the value zero. Trailing spaces are not required to fill out the lines to proper length.

Title line:

!...+....1....+....2....+....3....+....4....+....5....+....6
TT

where T is alphanumeric title data.

Initial origin coordinates:

```
!...+....1....+....2....+....3....+....4....+....5....+....6
OOOOOxxxxxxEEEEEEEE.EEENNNNNNN.NNNHHHHHHH.HHH
```

where O is alphameric data - the name of the origin,
 E is numeric data - the east coordinate in metres
 N is numeric data - the north coordinate in metres
 H is numeric data - the height coordinate in metres

It is not possible to enter these coordinates in feet.

Calibration of instruments:

```
!...+....1....+....2....+....3....+....4....+....5....+....6
CHANGExxxxxxxxxxxxDDDD.DCCCC.Cxxxxxx comments
```

where CHANGE is the keyword to change the instrument calibration.
 D is numeric data - the declination to be subtracted
 from input compass bearings.
 C is numeric data - the correction to be subtracted
 from input clino readings.

Normal input data:

```
!...+....1....+....2....+....3....+....4....+....5....+....6
FFFFFTTTTTTLLL.LBBBB.BCCCC.CVVV.VV comments
```

where F is alphameric data - the name of the station FROM
 T is alphameric data - the name of the station TO
 L is numeric data - the measured length in current
 units (default is metres)
 B is numeric data - the bearing in standard degrees
 C is numeric data - the clinometer reading in std.
 degrees (+ve up, -ve down)
 V is numeric data - a plumbed vertical component of
 the survey leg. This eliminates problems with
 clinometers having non-zero calibrations, when
 +90 readings get changed.

End-of-survey marker:

```
!...+....1....+....2....+....3....+....4....+....5....+....6
END xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx comments
```

where END is the keyword to terminate the input for this survey,
 the three blanks following the keyword must be present.

If the 'END' keyword occurs in a nested file, the user is asked whether to +
 halt input, or to resume input from the calling file.

The above should be enough for most simple surveys. Additional facilities are in-
 troduced by further keywords.

Change of current units:

```
!...+....1....+....2....+....3....+....4....+....5....+....6
INFREETxxxxxxxxxxxxxxxxxxxxxxxxxxxx comments
```

UNITS FEET
 UNITS METRIC/METRES

```
!...+....1....+....2....+....3....+....4....+....5....+....6
METRICxxxxxxxxxxxxxxxxxxxxxxxxxxxx comments
```

where INFREET changes the current units to feet and decimal
 fractions of feet. It is not proposed to support inches.
 METRIC returns the current units to metres.

Change of survey data style:

```
!...+....1....+....2....+....3....+....4....+....5....+....6
DIVINGssssssNNN.NNxxxxxxxxxxxxxxxx comments
```

```
!...+....1....+....2....+....3....+....4....+....5....+....6
NORMALLxxxxxxxxxxxxxxxxxxxxxxxxxxxx comments
```

where DIVING changes to the underwater style of data
 ssssss is the name of the starting survey station
 NNN.NN is its depth in current units

NORMAL changes back to conventional surveying style

STYLE DIVING / STYLE NORMAL

Input data for underwater survey:

```
!...+....1....+....2....+....3....+....4....+....5....+....6
FFFFFTTTTTTLLL.LBBBB.Bddddd.dDDDD.D comments
```

where F is alphameric data - the name of the station FROM
 T is alphameric data - the name of the station TO
 L is numeric data - the measured length in current
 units (default is metres)
 B is numeric data - the bearing in standard degrees
 d is the depth of the from station in current units
 D is the depth of the to station in current units

In most surveys, the depth of one station is the same as one of those in the pre-
 vious leg, and in most such cases, the field may be left blank. There are some r-
 estrictions, however, which are discussed in section 3.2.2 on page ??

New origin coordinates

```
!...+....1....+....2....+....3....+....4....+....5....+....6
ORIGINxxxxxxxxxxxxxxxxxxxxxxxxxxxx comments
OOOOOxxxxxxEEEEEEEE.EEENNNNNNN.NNNHHHHHHH.HHH
```

where ORIGIN is the keyword for new origin,
 O is alphameric data - the name of the origin,
 E is numeric data - the east coordinate in metres
 N is numeric data - the north coordinate in metres
 H is numeric data - the height coordinate in metres

The station name used for a new origin must not be the same as an existing sur-
 vey station name.

Unlike the initial origin, these coordinates are in the current units which may -
 be feet.

Nested data file call:

```
!...+....1....+....2....+....3....+....4....+....5....+....6
FILE fffffff
```

where fffffff is a valid filename on the current filing system. (The precise + length of this field varies for different filing systems, so is not restricted by the program itself. For this reason, nothing may follow the filename on the input line.)

The currently open file is closed, and the nested file opened for input. When the nested file runs out, input is resumed from the original file, with the line after the FILE keyword. The maximum nesting depth allowed is five nested files.

RESTRICTIONS

There are several restrictions on the input data to SU, few of which pose any problem to normal surveys. The total number of survey stations may not exceed 1.5k+ (1536). This is a restriction caused by lack of RAM, and is difficult to avoid + since the list of survey stations used needs to be kept in memory to allow a rapid search. 1.5k is about the limit on a standard model B machine, though the program could easily be modified to use sideways ram on a suitably equipped machine+. If the current linear search is replaced by a hashing algorithm, the search length should be reduced enough for the station list to be held on disc without being grossly slower. This should allow about 10000 stations to be processed, but + is still some distance in the future.

Survey stations can have names of up to six characters, which must not be the same as any of the keywords available. It is also sensible to restrict the characters used to those which can be printed on the installation in use, since it is useful for the user to be able to tell the stations apart! To maximise portability of data, it is recommended that station names be restricted to digits, upper case letters and a few common punctuation marks (which may include space). The author normally uses a two-letter, two-digit format where the first letter indicates the cave or part of the system, the second letter subdivides the survey into + convenient sized blocks, and the digits indicate sequence. Entrances, permanent + survey stations and other significant points have more distinctive names, typically of six characters, so that they stand out in a list of coordinates. The example input data file illustrates this usage.

While the program will quite happily detect and correctly process backsights, it is not capable of backtracking when a section of the system is 'surveyed out', + ie. the survey starts at an unknown point and proceeds towards an eventual connection with a known survey station. In this case, the conversion to vector form will be unaffected, but in the coordinate section of the printout, these stations+ will be flagged as 'UNDEFINED'. Input order is therefore restricted so that in + any survey leg, at least one of the survey stations will have been previously defined. It would be possible to overcome this restriction only by keeping all data in memory or by extensive disc searching, meaning small surveys or very slow processing. It was considered more valuable to be able to handle large surveys fairly quickly since it is normally fairly easy to resequence survey data to overcome this restriction.

Restrictions on diving data

To avoid the user needing to enter the depths at both ends of a survey leg, the + program will always remember the depth of the newly defined survey station on each leg, and use it if the same station is used in the next leg. This will only occur if the depth is not quoted in the new leg (or is zero). Thus, by explicitly+ quoting both depths, a change of depth gauge or water level between surveys is + allowed for. If a survey station was not newly defined in the previous leg, then+

its depth must be explicitly quoted, or will be assumed zero. A problem could arise in the unusual circumstance that a depth of zero needs to be explicitly quoted for a station that the program remembers as having a non-zero depth. The rather unsatisfactory ploy of giving .00001 is recommended here. In practice, the situation has never arisen with any survey processed by the author.

Running the program

The program and its support code must be in the current directory on the current drive (ie. use *DRIVE under DFS, *DIR under ADFS, NFS, ANFS). In the case of hierarchical filing systems, you will need directories O and U to hold the support code. The program is run by:

```
CHAIN "SU1"
```

There is an overlay handler (loaded from U.PROCFN) which will trap calls to missing procedures and load from "O.Pprocname" files as needed.

The program will ask for a path name to search for data files. In the case of a DFS system this will be the drive number. For hierarchical systems, it may be any arbitrary pathname leading to a directory containing subdirectories D, O, N, C and P, of which the D directory contains the data file(s). The data files need not be on the same disc or in the same directory as the program itself.

The system will next ask for the name of an input file. The format of this file is specified in "Entering Survey Data". The filename used is "D.filename", where "filename" is what you typed in.

Next, the program will ask for an output filename - if you just type <return>, it will use the same filename as for input, otherwise it will use the new filename. The program will create four new files:

```
O.filename      N.filename
C.filename      P.filename
```

The P. file is a copy of the printed coordinate listing, while the other files are generated for internal use by the SU-BBC system. If a hard copy device is available, it may either be enabled before the program is run, in which case all screen output will be printed, or it may be enabled in response to a question before the coordinate generation stage. If the printer is not used, the file "P.filename" may be printed after the program has finished, to obtain a list of the coordinates only.

OUTPUT

The screen output from the program is divided into three sections:

- * A reflection of the input data survey station names, with any commentary present.

- * A brief report on the survey statistics, including the number of survey stations, the number of these which are 'nodes', (ie. have other than two references - this information is relevant mainly for future versions of the program which may include loop closure algorithms), the plan and total length of the survey. An audible bleep announces the production of this section of output and the user is invited to specify whether output of the final section should be paged or continuous, and whether it is to be sent to a printer as well as the screen. The printer parameters (*FX5 and *FX6 etc.) are assumed to be already set up.

* The coordinates supplied for the initial origin, followed by, for each survey+ leg ; the coordinates of whichever of the two stations was not previously defin+ ed. If neither station was defined, then the word 'UNDEFINED' appears in place o+ f the coordinate data, while if both stations were already defined, a misclosure+ is printed. A station which has been set up via the ORIGIN keyword is preceded + by a "New origin" message. Note that whatever the units in which the data were e+ ntered, all output from the program is in metres.

GRAPHICS PROGRAMS

There are two graphics programs - both very crude at present, one for drawing a+ plan view, and one for an elevation. Each comes in two almost identical version+ s:

```
SUPLAN  SUELEV  use mode 0 for high quality monitors
SUPLAN1 SUELEV1 use mode 1 for lower quality TV's
```

The mode 1 versions can easily be altered to be suitable for drawing different p+ arts of the survey in different colours. For example, it is often helpful to hav+ e a surface survey drawn in a different colour from the underground one(s).

These programs do not need to be on the same disc (etc.) as the data files which+ they use, since once loaded, they remain memory resident unless overwritten by + new program material. The graphics routines use two of the files created by SU-B+ BC, viz. "O.filename" and "C.filename". Both programs require essentially the sa+ me input:

- * The filename for the data files
- * A scale, specified by giving the screen width in metres - the displayed heig+ ht is three quarters of the given width.
- * The compass bearing of the screen x-axis. This bearing will be from left to + right on the display.
- * X, Y and (for elevation only) Z coordinates in real space, of the point to+ be at the centre of the screen. These parameters are asked for by the programs + in a two line command window at the bottom of the display, and while fairly self+ -explanatory, do require a certain amount of mental agility to look at the right+ part of your survey. It is intended that the finally released graphics programs+ will be much improved in this respect. On the other hand, the author has got by+ for two years with these versions, written during a thunderstorm on an Austria + expedition and not modified since.

* A final question in some versions asks for a number of survey stations before+ changing the colour of the displayed line. This is to enable part of the survey+ (eg. a new extension) to be highlighted. It is obviously only applicable to m+ ode 1 displays.

Cassette tape distribution kit

The system is normally supplied on disc, since it requires a disc system to run.+ In a number of cases however, users may have 8" or 3" drives which the author d+ oesn't, or only a network filing system onto which floppy files can't be loaded + easily, and so the system can be supplied on tape. The standard distribution kit+ has all the files at 1200 baud on one side and 300 baud on the other side, and + is not guaranteed to be the latest release, because creating the tape takes fore+ ver.

The program, as supplied on tape, comes in 6 files:

1. OVERLAY
2. SU1
3. SETUP
4. INLINE
5. NODES
6. COORDS

To set the program up to run from disc, the overlay manager must be assembled an+ d stored on disc, together with the BASIC code in the other five files. It is on+ ly sensible to store a backup copy of the source code of the overlay manager on - disc too. The following commands will set things up correctly:

```
*TAPE
LOAD "OVERLAY"  :*DISC
SAVE "OVERLAY"  :REM does not need to be on same disc as SU
REM all the others need to be on the same disc:
RUN             :REM this will save overlay code to U.PROCFN
*TAPE
LOAD "SU1"      :*DISC
SAVE "SU1"      :*TAPE
LOAD "SETUP"    :*DISC
SAVE "O.Psetup" :*TAPE
LOAD "INLINE"   :*DISC
SAVE "O.Pinline":*TAPE
LOAD "NODES"    :*DISC
SAVE "O.Pnodes" :*TAPE
LOAD "COORDS"   :*DISC
SAVE "O.Pcoords":*TAPE
```

The program is now disc resident, and the description following applies to all v-
ersions of the program, whether supplied on tape or disc.